

FlowMeter Handoff Guide

A short guide for wiring, uploading the main program, changing settings, and reading the SD card.

Main program	Board	Last reviewed
Code/FlowMeterMain/FlowMeterMain.ino	Adafruit Feather M0	July 22, 2026

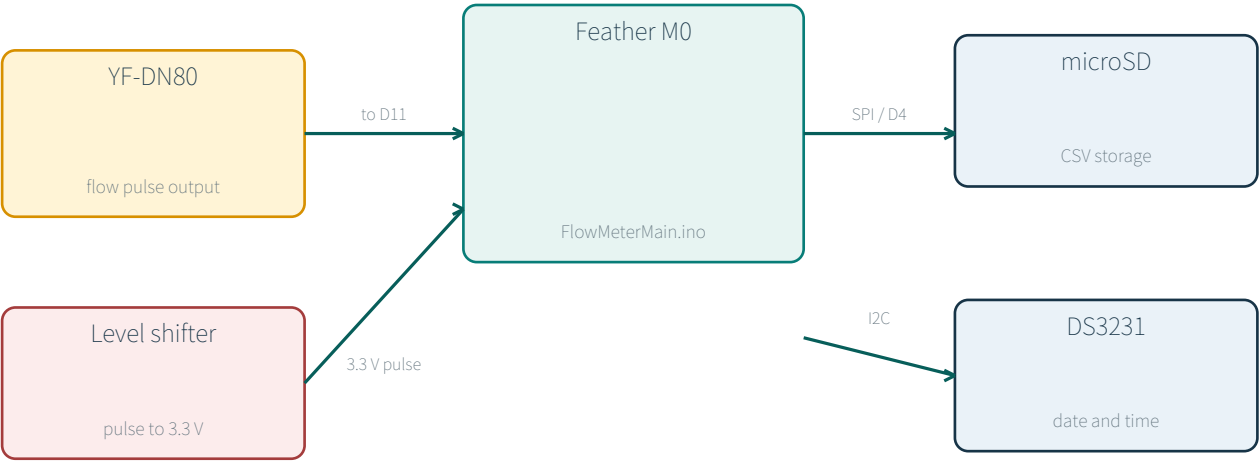
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Scope

This guide only covers the main logger program. It does not describe the separate test or calibration sketches.

Component Connections

Place the Feather M0, RTC, SD card, and level shifter in a dry enclosure. Install the YF-DN80 in the pipe with its flow arrow pointing in the actual flow direction.



Component wire	Connect to	Note
YF-DN80 signal (yellow)	Level shifter input	The sensor pulse must be shifted to 3.3 V logic.
Level shifter output	Feather D11	This is FLOW_PIN in the main program.
YF-DN80 power and ground	External supply + common GND	Use the voltage printed on the actual sensor. Connect sensor GND to Feather GND.
DS3231 VCC / GND	Feather 3V / GND	Powering the module at 3.3 V keeps I2C logic safe.
DS3231 SDA	Feather SDA	GPIO 20 on the Feather M0.
DS3231 SCL	Feather SCL	GPIO 21 on the Feather M0.
microSD CS	Feather D4	The main program uses SD_CS_PIN = 4.
microSD SPI	Feather SCK / MOSI / MISO	Only needed for an external SD module.

i Likely board configuration
D4 matches the built-in SD card on an Adafruit Feather M0 Adalogger. If that is the installed board, simply insert the microSD card; no external SPI wiring is needed.

! Electrical safety
The Feather M0 uses 3.3 V logic. Do not connect a 5 V sensor pulse directly to D11. Confirm the level-shifter output is no higher than 3.3 V before connecting the MCU.

Settings You Can Change

Open FlowMeterMain.ino and edit the values near the top of the file before uploading.

Setting	Current value	What it controls
FLOW_PIN	11	Flow sensor pulse input. Change only if the wire is moved to a different interrupt-capable pin.
SD_CS_PIN	4	SD card chip-select pin. Keep D4 for the Feather M0 Adalogger built-in SD card.
K_FACTOR_HZ_PER_L_MIN	0.45	Main flow conversion value. This directly changes the calculated flow rate.
CALIBRATION_SCALE	1.0	Optional multiplier. Normally leave at 1.0.
FLOW_OFFSET_L_MIN	0.0	Optional flow offset. Normally leave at 0.0.
MIN_VALID_FLOW_L_MIN	0.0	Readings below this value are changed to zero. Increase only if small noise appears when there is no flow.
SAMPLE_INTERVAL_MS	10000	Logging interval in milliseconds. 10000 means one CSV row every 10 seconds.
FORCE_SET_RTC_TO_COMPILE_TIME	false	Temporarily set true to set the RTC from the computer's compile time.

Flow calculation

```
frequency_hz = pulses / elapsed_seconds
flow_l_min = frequency_hz / K_FACTOR_HZ_PER_L_MIN
flow_l_min = flow_l_min * CALIBRATION_SCALE + FLOW_OFFSET_L_MIN
```



Current K factor

The main program currently uses 0.45, while the YF-DN80 nominal relationship is commonly $F = 0.5 \times Q$. Keep 0.45 only if it came from the installed sensor's field calibration; otherwise verify it with a measured-volume test.

Setting the RTC

- Set FORCE_SET_RTC_TO_COMPILE_TIME to true.
- Compile and upload once, then check the printed RTC time.
- Change the setting back to false and upload again for normal operation.

Upload and Start Logging

Use the Arduino IDE to compile and upload the main program to the Feather M0.

Arduino IDE setup

Item	Required setting
Board package	Install Adafruit SAMD Boards in Boards Manager.
Board	Select Adafruit Feather M0.
Library	Install RTCLib by Adafruit. SPI, SD, and Wire are standard Arduino libraries.
Serial Monitor	115200 baud.

Upload steps

Step	Action
1	Connect the Feather M0 to the computer by USB.
2	Open Code/FlowMeterMain/FlowMeterMain.ino in Arduino IDE.
3	Select Adafruit Feather M0 and the correct COM port.
4	Check the settings on the previous page, then click Upload.
5	Open Serial Monitor at 115200 baud.
6	Confirm that the RTC is found, the SD card initializes, and the message Logger ready appears.

Expected startup messages

```
Starting YF-DN80 + DS3231 + SD logger...
RTC startup time: YYYY-MM-DD HH:MM:SS
Initializing SD card...
SD card initialized.
Logger ready. Writing to: /YYYYMMDD/HMMSS.CSV
```



If upload fails
Recheck the selected board and COM port. On some Feather M0 boards, double-pressing Reset starts the bootloader and creates a temporary upload port.



Before normal logging
Insert the SD card before powering the logger. If the RTC or SD card cannot initialize, the program stops and does not create data rows.

Writing and Reading Data

The logger creates one new CSV file every time it powers on. It writes and closes the file after each sample.

Card preparation

- Use a microSD card formatted as FAT16 or FAT32.
- Insert the card before the Feather M0 starts.
- Do not remove the card while the board is powered and logging.

File location

/YYYYMMDD/HHMMSS.CSV
Example: /20260722/143015.CSV

CSV columns

Column	Meaning
timestamp	Date and time from the DS3231 RTC.
session_elapsed_seconds	Seconds since the board restarted.
sample_number	Row number for the current power-on session.
pulses	Raw flow-sensor pulse count during the sample interval.
frequency_hz	Pulse frequency.
flow_rate_l_min	Calculated flow in liters per minute.
total_volume_l	Accumulated volume for the current power-on session.

How to read the card

Step	Action
1	Stop the water flow and turn off power to the logger.
2	Remove the microSD card and insert it into a computer.
3	Open the folder named with the date, then open the CSV file named with the start time.
4	Open the CSV in Excel, Google Sheets, or a text editor. Keep a copy of the original file.



Important
total_volume_l resets to zero whenever the MCU restarts. It is a session total, not a permanent lifetime total.

Basic Troubleshooting

Start with the first error printed in Serial Monitor.

Problem	What to check
No pulses / zero flow	Check water flow, common ground, level shifter, and the D11 signal wire.
Small flow when water is off	Check electrical noise and grounding first. If needed, increase MIN_VALID_FLOW_L_MIN slightly.
Flow is consistently high or low	Verify K_FACTOR_HZ_PER_L_MIN. Do not change K and CALIBRATION_SCALE at the same time.
RTC not found	Check 3V, GND, SDA, and SCL wiring.
Wrong date or time	Use the one-time FORCE_SET_RTC_TO_COMPILE_TIME procedure, then return it to false.
SD initialization failed	Check that the card is inserted, FAT16/FAT32 formatted, and using CS D4.
CSV file is missing	Check the RTC date folder, Serial errors, card format, and available space.

Final handoff checklist

- ☐ The installed board and SD configuration are identified.
- ☐ The sensor pulse reaches D11 through a 3.3 V level shifter.
- ☐ RTC time is correct and FORCE_SET_RTC_TO_COMPILE_TIME is false.
- ☐ The correct K factor and 10-second logging interval are confirmed.
- ☐ A CSV file has been created and opened successfully on a computer.

 Main source file
Code/FlowMeterMain/FlowMeterMain.ino